



Programming Autonomous Robots
COSC2781 (PG) | COSC2814 (UG) | Semester 1 2026
Assignment | Robotics Project (DRAFT)

Assessment Type	Group Assessment. Clarifications/updates may be made via announcements/relevant discussion forums.
Project Selection	No later than, Thursday Workshop in Week 9
Progress Update	During Workshops in Week 12
Demonstration	During Week 15 (by appointment)
Report Due Date	23.59pm, Sunday June 21st 2026 (Week 15)
Weight	50% of the final course mark
Submission	Online via Canvas. Submission instructions are provided on Canvas.
Learning Outcomes	This assignment contributes to CLOs: 1 - 5

1 Overview

In the project you will research how to solve a problem or algorithm in autonomous robotics, that can be solved or demonstrated on one of the AI Innovation Lab or VXLab robot platforms, such as the ROSBot Pro 3.0 (and Pro 2.0), Softbank Nao V6, Booster K1, Fourier GR1-Pro, Panther, Tiago, UFactory xARM6, or UR5e Arm. You will implement at least one solution to this problem, and conduct a live demonstration of the effectiveness of your implementation on the physical robot platform.

You will need to show in-depth capabilities in at least one of the core topics about autonomous robotics that are discussed in this course. You will need to upon your course knowledge, by diving into deeply into one or more of these topics. You will need to conduct research into state-of-the-art approaches, and adapt these methods for your project. You will be assessed on your research, the implementation of your project, and the analysis and evaluation of the capabilities of the software you develop. You may use simulation to help develop and test your software, however, your work must be evaluated on a physical robot. You will present your final work by a live demonstration, along with a report detailing your research, analysis and evaluation.

This specification provides suggested starting point for your project that you may choose from. The exact scope will be discussed and determined, with the course coordinator, depended on the make-up of the group. You may also negotiate a project with the course coordinator. All projects need to be chosen, and with the project scope determined, by the Thursday Workshop in Week 9.

Aspects of this assessment, and the rubric, are different for UG and PG students. Carefully check the requirement for your level of study. The summary of differences is:

- UG students must exhibit the depth and breath of their technical capability in designing and implementing autonomous robot software. This may vary for each project but could include:
 - Implementation of, and adherence to, a specific robot software architecture.

- Implementation of ROS2 software required to enable robot functionality, outside of the AI autonomous behaviour.
- Implementation of a broad ROS2 autonomous software stack, drawing on a diverse range of topics across motion, vision, navigation, mapping, and task planning.
- Implementation and comparison of multiple methods or algorithms for tackling the project.
- PG students must design, prepare, and conduct experiments to evaluate their project implementation, where the experiment design clearly shows the strengths and limitations of their work. That is, PG students will need to design an experiment that would be suitable for inclusion in a published work/paper on their project that would prove to an external reviewer the quality of their work. PG students will need to carefully consider how to evaluate and analyse their work, synthesising their knowledge gained in this course in the practical implementation of algorithms for autonomous robots.

1.1 Group Work

You are required to work groups of 3-4 students. Groups may be formed from both UG and PG students. This is because of:

1. The limited availability of the physical robots
2. To minimise wear-and-tear and prevent over-use of the robots
3. Safety, in ensuring appropriate monitoring of the robots during autonomous operation

Note that unlike in the first assessment, in the project there are specific grades for your individual contribution to your group's work. In your report you will need to record the individual contributions of each group member. In the event of group-work issues, individual grades may be allocated. Therefore, we advise that your group uses suitable group work tools, such as MS Teams, Task Planner/Trello, and a Git Repository. We also recommend that you work closely together on the assignment, developing the code together, so that each member of the team has a good understanding of how all of the parts of your final software system work.

1.2 Learning Outcomes

This assessment relates to the following course learning outcomes:

- CLO1** Discuss and Critically Analyse a variety of software architectures and algorithms for solving typical problems in the context of autonomous robot systems; Discuss and Critically Analyse the strengths and limitations of these architectures and algorithms.
- CLO2** Discuss and Critically Analyse the challenges of designing and developing software for a variety of robot systems of different complexities, including noise, uncertainty, and computational power.
- CLO3** Research, Discuss, and Use new and novel algorithms for solving problems with autonomous robot systems.
- CLO4** Use pre-existing robot software to solve common problems on simulated and real-world robots; Develop and Implement new algorithms and software for solving problems on simulated and real-world robots; Integrate this software in the ROS framework.
- CLO5** Develop skills for further self-directed learning in the general context of software, algorithms, and architectures for autonomous robot systems; Adapt experience and knowledge to and from other computer sciences contexts such as artificial intelligence, machine learning, and software design.

1.3 Use of Artificial Intelligence (AI) Tools in this Assessment

Perhaps compared to other courses, the use of AI tools and Generative AI in this course is generally quite open. This being noted, it remains the case that ***the majority of your work in this assessment should be your own work***, and not plagiarised from other sources, or sourced from the use of Artificial Intelligence (AI) tools. Therefore, the use of AI tools are restricted in certain ways for this assessment task.

In this assessment task, you may use AI tools to support you in developing ROS2 boilerplate software, generating ideas, planning, and/or drafting. You may find it helpful to use AI Tools for:

1. Boiler-plate ROS code, such as node templates
2. Boiler-plate ROS package and configuration files
3. Ensuring ROS syntax is correct, such as topic types
4. Debugging code compilation errors

We also suggest reviewing the Course AI Policy on Canvas for hazards and pitfalls in using AI tools in this assessment. Any use of such tools must be acknowledged and referenced. Work that is significantly produced by

AI tools, or where AI tools are used to complete this assessment without attribution may result in an allegation of academic misconduct.

For the purposes of openness, transparency, and academic integrity, you must submit a log of your use of generative AI tools during this assessment. This also means you should think about how you track your AI use. Each tool has a way of exporting your interaction history. The below links describe how to export your logs from different AI tools:

- ChatGPT
- Claude
- Gemini

Suspected use of AI tools without a log may result in grade penalises and/or misconduct proceedings.

2 Assessment Details

The purpose of the project is to enable a robot to autonomously complete a task. Each project is scoped differently and the requirements of each project are laid out in Section 4. All of the projects share the features described below, which you will be required to address or conduct through the course of your project.

2.1 Investigation and Research

You will need to investigate and research algorithms/techniques related to your project. You will need to implement and adapt these existing techniques to your particular needs. In your report, you must record the findings of your investigate, and cite research relevant to your work. You must fully cite any references that you have used within your research, implementation, analysis and/or evaluation.

2.2 Original Implementation

You must complete an **original implementation** of at least one significant element of the autonomous software for the robot as part of your project. That is, your work cannot exist entirely of off-the-shelf software, such as existing ROS2 packages. The scope of this original implementation is defined within the project definition and negotiation with the course coordinator. This original implementation may:

- Significantly modify or adapt existing techniques and algorithms for practical use on a robot.
- Be a novel implementation of your own creation.

Your report should describe your original implementation.

2.3 Robot Software Architecture

You must consider and choose an appropriate robot software architecture (such as SPA, Subsumption, or a three-tier architecture) for implementing your solution to the project. You will need to describe and justify your choice of robot software architecture in your report, and show that your project adheres to this architecture.

2.4 Autonomy

Your software should be fully autonomous, unless the scope of your project says otherwise.

2.5 Open-ended Exploration

Each project is designed to be open-ended, allowing for multiple avenues of exploration. A purpose of the project is for students to synthesis their knowledge across the course. More complex solutions to the projects displaying more complex autonomous software will receive higher grades compared to trivial or simplified approaches.

2.6 Demonstration & Analysis

You (all groups) must complete a demonstration your autonomous software at completing the challenges of your particular project. You (all groups) must also complete an analysis of their software, presented in your report. This analysis should:

- Highlight the strengths and capabilities of your work.
- Identify limitations or weakness of your work.

You will need to collect evidence to support what the capability of your project.

2.7 Report

You must write a report to accompany your final demonstration. Details are provided in Section 6.

2.8 Activity Risk Assessment and Equipment Risk Assessments

All projects must complete an Activity Risk Assessment (ARA), before development activities or use of the robot may commence. The ARA must be approved by the course coordinator before you can commence work on your project. The ARA template is available on Canvas for the project.

Some projects may also require completing additional induction, along with reviewing and signing associated Equipment Risk Assessments.

2.9 Working in other Labs

For some projects, you may need to work in other locations and labs, including the VXLab (Ground floor, Building 91), or Bundoora spaces. This will require coordinating with staff who run these facilities, completing necessary inductions, and complying with all relevant safety requirements.

For the VXLab, you will need to complete the VXLab induction¹.

For Bundoora, induction links will be provided.

3 Separate PG and UG requirements

The requirements for PG and UG students, to satisfy these different levels of study are described in this section. Please review this for your level of study.

3.1 ROS2 Infrastructure & Robot Software Architecture (UG)

For working with some robot platforms, the software infrastructure is not prepared to the same degree as with the ROSBot's. Some platforms may also use older ROS2 systems, such as Humble. This is often due to the robots being setup for research purposes. Therefore, some projects will require UG students to implement additional infrastructure to enable working with the platform.

This may also include infrastructure that is implemented to ensure that the project is implemented according to the chosen robot software architecture. This can include implementing data structures and ROS2 nodes that enforce the flow of sensing information and behaviour generation.

3.2 Extended/Additional Algorithm Implementation (UG)

All projects have different algorithms/approaches that can be used to complete each project. UG students will be expected to implement multiple solutions to their project, if the infrastructure implementation is not sufficiently significant (or were not attempted). Unless otherwise noted in the project description, this will be discussed with the course coordinator on a per-project basis.

3.3 Experimental Design and Evaluation (PG)

As studies extensively in this course, every algorithm has strengths and weaknesses. While every project needs to present a discussion of these strengths and weakness, PG students must take this a step further, and design an experiment that explicitly highlights both the strengths of their work, and in particular, the limitations.

That is, PG students will need to consider their algorithm, to design an experiment where evidence can be collected to specifically showcase the scope of their implementation. This is more than just recording the robot in action. The standard of evaluation, is to collect evidence, including repeatable statistical measures, that would be sufficient for a published article.

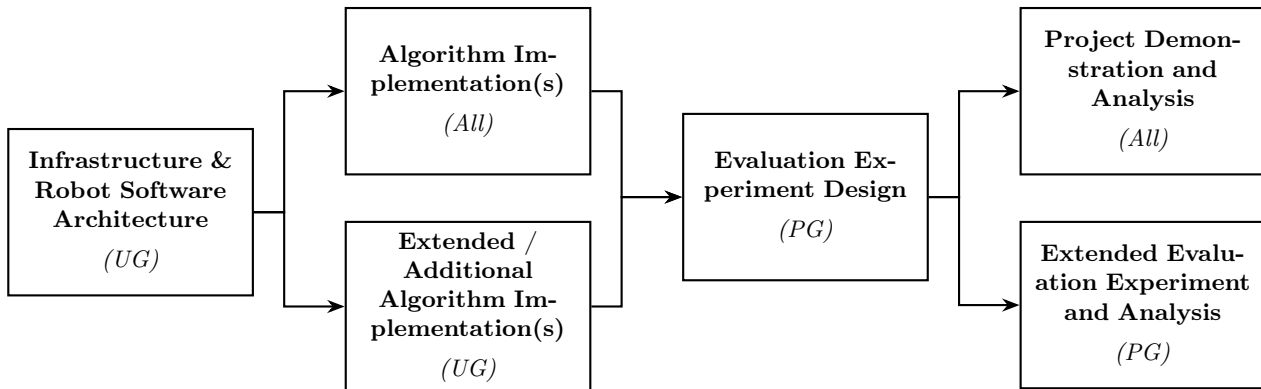
PG students will need to propose their experiment methodology during the progress update (During Workshops in Week 12). Therefore, PG students should also consider the time frame of when to complete their implementation(s) to allow sufficient time to successfully conduct experiments towards the end of their project.

¹<https://forms.office.com/r/YMcPnDuCQv>

3.4 Grading of PG/UG students in the same group

In this course we build a community that actively works with and supports each other, which is independent of the degree any student is studying. Therefore, it would be against the motivation of this course to isolate students to groups based on their level of study. Instead, we wish to leverage the strengths of both UG and PG levels of study, while also ensuring each student can show their capability relative to their level of study.

The following diagram gives an approximation of how the PG/UG elements can integrate into a unified project. This is not to say that elements are 100% exclusive to the groups, depending on the project, however, these are the areas in which each student is expected to make the majority of their contributions to the assessment.



For grading, each student will be graded according to the rubric for their level of study. Generally, all students in the group will receive the same grade in rubric components that overlap, and all students of the same study level in the group will receive the same grades for their level of study.

4 Projects

Below are suggested projects. The scope of each project is briefly described. You will need to further discuss and negotiate the scope of your project work with the course coordinator, and agree on the final deliverable. The scope may, by negotiation, change throughout the project. The agreed project scope, and any changes, must be documented in your final report.

4.1 Human Safe Animal behaviours with Unitree Go2

This project is not yet confirmed, and subject to interest from the course.

This project is to enable the Go2 robot from Unitree to simulate the actions of a pet, leveraging its dog-like form, within an open space with multiple people. The Go2 comes with a selection of behaviours which can be individually triggered to given the robot the appearance of a dog. This purpose of this project is to enable the robot walk safely about an open space, without running into other people or static obstacles, intermittently stopping and activating pre-programmed routines that exhibit the behaviours of an dog. Importantly, the behaviour of the robot should ensure the absolute safety of both the robot, and any obstacles within the space. This particularly includes maintaining safe distances from people, and stopping (lying down) if any person approaches too closely.

The scope of this project is to:

1. Devise a suitable behaviour system to move about an open space and trigger various pre-programmed behaviours.
2. Determine appropriate sensors and algorithm to monitor the surroundings of the robot.
3. Ensure the robot never collides with obstacles or people.
4. Ensure the robot enters a safe mode when people get too close to the robot.

This project will be conducted in collaborate with Kate Fox from Biomedical Engineering. The robot is not yet setup and enabled for autonomous operation. Thus the scope of this project includes:

- Investigating the Unitree ROS2 interface and devise infrastructure to control the Go2 through ROS2.
- Investigate the Unitree SDK Sports Services for how to trigger appropriate behaviours from the SDK.
- Implement a Robot Software Architecture for your behaviour and movement system.

UG students will be expected to significantly contribute to the development of the ROS2 infrastructure, and creation of a suitable Robot Software Architecture. PG students will need to carefully consider how to evaluate both the safety of the final system, and overall quality of the system.

4.2 RoboCup Soccer

This project is to enable the Booster K1 robot's to compete in the RoboCup HSL 2026 competition. See the HSL Rules 2026² for details. The competition includes both the soccer competition proper, and technical challenges. The rules define the scope of deliverable and requirements for the completion of the project. In negotiation with the course coordinator, this project will select a component of the rules to complete for the project. The following aspects are available for the the scope of the project:

- Localisation of the robot on the Soccer field.
- Robot vision and detection of field features, ball, and other robots.
- Motion for reliable movement and kicking.

For this project you will need to use the existing RedBackBots RoboCup codebase. This codebase already contains a defined robot software architecture, which is not permitted to be modified to prevent interference with preparations for the RoboCup competition held in July 2026. Therefore, UG students will need to explore multiple solutions to their problem.

You must demonstrate your final code on the Booster K1 robot under competition conditions, and PG students must consider how to effectively design an experiment for competition conditions.

4.3 RoboCup@Home Human-Robot Interactions tasks for Tiago

This project is to support founding a RoboCup@Home team at RMIT, using Tiago. This RoboCup league focuses on HRI tasks for domestic service robots. For this project, you will need to implement software to complete a reasonable representation of the “*Carry My Luggage*” challenge of the RoboCup@Home 2024 ruleset³. You should aim to comply with as much of the rules as possible, including the “Deus Ex Machina” contingencies. However, you will need to adapt the challenge task to the constraints of the VXLab, such as defining suitable ‘car’ and ‘delivery’ locations.

The scope of this project is to:

1. Learn how to work with Tiago.
2. Devise a suitable version of the “*Carry My Luggage*” challenge.
3. Determine the most appropriate Robot Software Architecture.
4. Choose and implement a suitable algorithms for enabling Tiago to complete the challenge.
5. Demonstrate your solution on Tiago.

Note that at present the Tiago robot in the VXLab uses ROS Noetic, thus as part of this project you will need to learn the differences between ROS2 and ROS1.

UG students will be expected to enforce an explicit robot software architecture suitable for this task. Extended implementations could be explored, depending on the direction of the project. PG students should consider competition conditions in designing their experiment

4.4 Augmented Reality for Explainable AI with the ROSBot

This project is to visual the actions and state the ROSBot Pro 3.0 (and Pro 2.0) in Augmented Reality using MetaQuest. This project ties into an research in robotics (and AI in general), known as Explainable AI (XAI). XAI is about trying to communicate and explain the action of an autonomous system to humans in an interpretable and understandable manner. A key challenge to this project will be integrating information from the reference frame of the robot into the MetaQuest. A second key challenge is studying how people unfamiliar with autonomous robots respond to your XAI system.

The scope of this project is to:

1. Determine the state and action information of the ROSBot Pro 3.0 (and Pro 2.0) to visualise
2. Choose a suitable task for the ROSBot Pro 3.0 (and Pro 2.0) to complete that for which visualising this task in AR would improve the explainability of the robot's actions to human observers.
3. Setup the AR software infrastructure libraries for the AI Innovation Lab.

²<https://github.com/RoboCup-HumanoidSoccerLeague/HSL-Rules/>

³<https://github.com/RoboCupAtHome/RuleBook/releases/download/2024.2/rulebook.pdf>

4. Integrate ROS2 software and the MetaQuest software for communication between the platforms.
5. Align the localisation of the ROSBot Pro 3.0 (and Pro 2.0) and the MetaQuest position in a room.
6. Render state and action information of the ROSBot Pro 3.0 (and Pro 2.0) in AR.
7. Write minimal ROSBot Pro 3.0 (and Pro 2.0) behaviours of the chosen task to clearly demonstrate the effectiveness of AR towards Explainable AI in autonomous robotics.
8. Demonstrate your system with the ROSBot Pro 3.0 (and Pro 2.0) and MetaQuest.

This project has considerable infrastructure requirements for UG students to create the MetaQuest-to-ROSBot interface. UG students should additionally consider the ability to enable future projects to continue on their activities. PG students will need to consider how their experiment can show the quality of their AR explanations.

4.5 Deep Reinforcement Learning of Robot Control

This project is to use online Deep Reinforcement Learning algorithms that learn how to control an autonomous robot. This project can be conducted with any of the AIIL or VXLab robots, including Panther, Fourier GR1-Pro, ROSBots, the UFactory xArm6, the UR5e, or Tiago. If using Panther, this project can be learning control of tasks with the UR5e mounted on the robot. In this project, you will work with an existing algorithms for online Deep Reinforcement Learning to investigate one method for control. The key to this project will be implementing an algorithm and conducting real-world experiments showcasing the learning and control. To complete this project you will need to define a suitable task.

You will need to choose which online Deep Reinforcement Learning algorithm such as:

- DREAMER (or variants)
- SAC (or variants)
- TD3 (or variants)
- Other algorithms of your choosing

The scope of this project is to:

1. Choose and Research a Deep Reinforcement Learning algorithm
2. Define a suitable control task for learning
3. Implement the chosen algorithm within the existing codebase(s).
4. Devise a suitable control or navigation task.
5. Conduct real-world experiments to evaluate your chosen algorithm.
6. Demonstrate your final trained controller.

Example tasks could include:

1. Learning to kick an AFL ball with Booster K1.
2. Learning to move different objects into a container with a robot arm.
3. Learning to ‘catch’ a moving ball with a robot arm.
4. Learning to pick-up small objects with the Panther arm.

The adoption of a robot software architecture in reinforcement learning contexts is challenging. Instead, this project has infrastructure requirements to create a learning framework, sufficient for UG students. Alternatively UG students may explore Sim2Real learning or the use of multiple algorithms. PG students will need to consider experiment conditions to explore the scope of reinforcement learning.

4.6 LLM Empowered Conversational and Sociable Robot

This project is to enable a humanoid robots, such as the Softbank Nao V6, Booster K1, or Fourier GR1-Pro robot to be a sociable robot, capable of holding a conversation with a person, where the conversation is backed by a LLM interface (such as Gemma4). For example, the robot might be able a greeter for RMIT labs when visitors arrive. However, the sociable robot must be more than just an microphone and speaker for an LLM. For social robots to be accepted in human environments, they need to move and react in ‘natural’ (or at least acceptable) ways. This include body movement and gestures that shows a presence within the social setting. LLMs can be enabled to not just provide textual output, but also select from a library of actions in conjunction with spoken output. Thus, this project is to build a system to interface a humanoid robot with an LLM to reply to spoken conversation, while exhibiting ‘natural’ movement in response to the environment.

To complete this project, you will need to:

- Research how to enable LLMs to both produce textual output and select from a library of control based on inputs
- Select algorithms to provide input to the sociable robot interface.

- Select a the library of algorithms and/or functions to enable the movement and control of the robot.
- Determine scope of conversation topics.
- Devise a suitable persona for the LLM.
- Develop the interface between the robot software, and the LLM.

This project may be conducted in collaboration with the RMIT RACE Hub, who can provide access to AWS Services to power the LLM backing of the robot.

UG students will be expected to significantly contribute to the infrastructure to support this project. PG students will need to consider how to measure the quality and sociability of the robot, which is non-trivial within the setting of social robots.

4.7 Navigation to Simulate Macadamia/Chestnut Farm Residual Harvesting

This project is to enable the ROSBot Pro (or Panther robot) to complete a simulated outdoor navigation task in an environment similar to macadamia/chestnut farms. A representative environment of an Orchard for a macadamia/chestnut farm is a field with multiple rows of trees, and the task is to drive the robot around the trees to collect nuts as if it was towing a harvester. This project combines challenges in navigation, object detection, and path planning. The key challenge is to create an full autonomous system, and will require careful consideration of the Robot Software Architecture, and coordination of the different software elements.

For this project you will need to:

1. Devise a simulated navigation task, including a suitable representation of a macadamia environment, suitable representation of rows of trees, starting location of the robot, and location of return of the robot.
2. Determine the most appropriate Robot Software Architecture.
3. Choose and implement navigation and mapping algorithm(s).
4. Choose and implement object recognition algorithm(s).
5. Choose and implement a suitable planning and coordination algorithm.
6. Demonstrate your software architecture and functionality on the Panther robot.

You may leverage a simulation environment on TheConstruct to help with experiment and testing, however, your final demonstration and evaluation must be on a real robot.

UG students will be expected to implement and enforce an explicit robot software architecture suitable for this task as part of creating the software infrastructure for the chosen robot. PG students should consider how to best simulate macadamia/chestnut conditions within the confines of our lab space(s).

4.8 Pick-and-Place tasks with Cobot UR5e Arm

This project is to enable the Cobot UR5e Arm, or UFactory xArm6 arm, to complete a pick-up and placement tasks in a fixed environment. This projects combined challenges of object visual recognition, manipulation, and task-planning. The task could be a pick-and-place task such as block-stacking, shelf stacking, item bin sorting, or a 2D player game. However, the type of task you choose will need to include a task-planning component. The key challenge is to create a fully autonomous system, from low-level control to high-level task planning, and will require careful consideration of the Robot Software Architecture.

For this project you will need to:

1. Devise an appropriate pick-and-place task, that includes a task-planning component.
2. Configure the software stack of the Cobot UR5e arm.
3. Determine the most appropriate Robot Software Architecture.
4. Choose and implement appropriate vision object recognition algorithm(s).
5. Choose and implement appropriate manipulation algorithm(s).
6. Choose and implement a suitable task-planning algorithm.
7. Demonstrate your software on the Cobot UR5e arm.

Example tasks could include:

1. ‘Catching’ a moving ball with a robot arm.
2. Playing a ‘2D’ game of Connect-4.
3. Stacking blocks of different shapes.

Each arm will require some software infrastructure to be created. UG students will also be expected to implement and enforce an explicit robot software architecture suitable for this task as part of creating the software infrastructure. PG students should consider how to best show the capability of the task execution, especially on how to demonstrate limitations of their project.

4.9 Multi-ROSBot Game

This project is to enable multiple robots to independently and autonomously play a game like ‘follow-the-leader’, or ‘tag’ within a pre-defined space. You should use multiple ROSBot Pro 3.0 (and Pro 2.0) robots (at least three). You will need to design your software to allow each robot to be as independent as possible, with limited coordination by a ‘centralised’ controller (likely running on your own device). You will need to consider how to avoid collisions with both other robots and environment obstacles.

A challenge of this project is ROS2 how to enable the multi-robot game. Our software setup for the ROSBot’s does not support multiple robots directly communicating together. You are not permitted to change this configuration. That is, each robot should be an independent player of the game, much like individual humans playing similar games. Instead, you may have a single coordinator node, that can share information between multiple nodes. This coordinator could be put onto a robot, or one an external device. You will need to consider how to write your algorithm(s) for independent robot action, with limited coordination from a central system.

You will need to determine how to manage the coordinator node. We recommend using ROS2 domain bridge to enable sending/receiving very limited information between each robot and the coordinator. You will need to consider what information to send.

The scope of this project is to:

1. Define a suitable swarm task.
2. Research swarming algorithms.
3. Research ROS2 configuration requirements to support swarms of robots.
4. Implement swarm software to coordinate the actions of multiple robots to complete the swarm tasks while avoiding collisions.
5. Demonstrate the swarm behaviours in simulation and/or on the real ROSBot Pro 3.0 (and Pro 2.0) robots.

This project has scope for UG students in setting up both the infrastructure for coordinating the robots and enforcing suitable robot software architectures on every robot. PG students will need to consider the complexities of designing a repeatable experiment with multiple robots.

4.10 Person Recognition and Tracking

This project is to enable the Panther to recognise and respond to the presence of 4 specific people that the robot has been trained to recognise, and then follow one of the recognised people as they move. The recognition and tracking task will be completed within an indoor environment, but is not confined to the space of the AI Innovation Lab. That is, could be conducted within building environments at RMIT, such as Building 14 spaces, and classroom spaces.

The scope of this project is to:

1. Create original software for to enable the Panther to recognise at least 4 unique individuals.
2. Determine appropriate sensors and data by which to train and recognise the individuals.
3. Write behaviours for the Panther to track and follow the individual that it has recognised in an indoor environment.
4. Determine appropriate sensors and data by which to follow a person.
5. Research potential software features of the ZED camera to complete the above scope.
6. Demonstrate your solution on Panther.

You will need to research methods for efficient real-time person recognition and tracking, and how to most effectively leverage the Panther sensors. Person recognition could be achieved by:

- Facial recognition
- Recognising body shape and size
- Voice recognition

UG students will be expected to implement and enforce an explicit robot software architecture suitable for this task as part of creating the software infrastructure for the chosen robot. PG students should consider how to best evaluate the robot’s person recognition, including the training process and successful recognition of multiple different people.

4.11 ROSBot Package Pickup and Delivery

This project is to devise and implement a single ROSBot dynamic pickup and delivery task, where a ROSBot must repeatedly search for simulated package pickup locations, and deliver the package(s) to simulated delivery

locations. The challenges in this project are based around search, navigation, vision, and task planning. You will need to devise a suitable version of this delivery challenge, which is an extension of the below minimal requirements. A key challenge in this project is maximising the number of packages that can be delivered within a limited timeframe. This will require careful thought into both how you devise a suitable delivery challenge, and the task planning algorithms that you implement to manage package pickup and delivery.

Minimally, the ROSBot will need to:

- Have a rigid 'pointer device' temporarily attached to the robot, that can ideally be seen in the camera.
- Search an unstructured and dynamic environment for one or more Aruco markers.
- At least one marker will denote a "pickup" location, from where the ROSBot must visit and 'touch' the marker to pickup a unique package.
- At least one marker will denote a unique "delivery" location for the package. The ROSBot must visit and 'touch' the marker to deliver the package.
- 'Carry' multiple packages at once, that is, visit multiple pick-up locations and deliver the packages to multiple delivery locations.
- Track the package carrying and delivery status.
- Continue to operate for a 15 minute period delivering as many packages as possible.
- Produce an accurate map and log of all actions of the ROSBot over the operation.

During the run:

- There will always be at least one pickup-delivery Aruco marker pair.
- The pickup and delivery Aruco markers may be added, removed, and/or repositioned.
- Dynamic obstacles may move about the unstructured environment, including other robots and people.

You may use mecanum (omni-directional) wheels on the ROSBot, and will likely find these useful for implementing control logic to touch the Aruco markers. If taking this approach, you will need to speak with the course staff about the physical and software setup to enable.

For this project you will need to:

1. Source unique Aruco markers to use.
2. Define a mapping of Aruco markers to unique pickup and delivery points.
3. Source an appropriate pointer device and attachment method.
4. Develop suitable robot visual algorithms to identify Aruco markers.
5. Research and develop suitable navigation and behaviour coordination logic.
6. Determine the most appropriate form(s) in which to provide a map and log of the ROSBot's actions during its operation.
7. Research and develop suitable task planning logic to manage package deliveries.
8. Demonstrate your solution on the ROSBot.

4.12 Negotiated Project

If you have your own ideas for a project in this course, then you may **propose and negotiate** a project with the course coordinator. This proposal and negotiation will need to clearly define the scope of the project and the expected deliverables for the demonstrations.

You should begin this discussion as soon as possible. The scope of the negotiated project must be finalised by no later than the Thursday Workshop in Week 9.

5 Demonstrations

You will conduct 2 demonstrations of your work throughout the course of the project. The first demonstration will be a progress update during Week 12 classes. The second will be the final demonstration of your project work, held by appointment during Week 15.

5.1 Week 12 Progress Demonstrations

During Workshops in Week 12 you will hold a **progress update** on the work your group has completed.

Your progress update should demonstrate and discuss:

- That the project is on-track according to the project scope
- Components for a minimally viable solution to the project

Your progress update should discuss the timeline of your completion of the project. This should include:

- A feasible plan for the completion of the project
- Improvement of the project beyond the minimally viable solution
- Intended deliverables to be demonstrated during the final demonstration
- (UG) A feasible plan and improvements of the project for extended infrastructure, architecture, and/or implementation
- (PG) Proposed experimental evaluation of the project

As part of this update you will submit on Canvas:

- A complete copy of your current software
- Your plan for completion of the project
- A record of evidence of your individual work

5.2 Final Demonstrations



Note that this demonstration will happen *before* you make your final submission.

The final assessment of your project will be conducted via a live demonstration in During Week 15 (by appointment). The demonstration will happen *before* you make your final submission so that you have time *after* the demonstration to complete your final report. Your final demonstration should:

- Show that the project meets the scoped requirements.
- Distinguish your work's original implementation to from existing literature, software and/or dependencies.
- Showcase the capabilities of your implementation.
- (UG) Discuss/showcase your extended infrastructure, architecture, and/or implementation.
- (PG) Discuss/showcase the results of your experimental evaluation. You should aim to demonstrate a subset of these results.
- Be well-prepared with minimal time lost for setup, configuration, and restarts. You may supplement your demonstration by additional material or recordings.

6 Report

You are required to write a report that:

- Describes the Methodology of your development and work.
- (UG) Describe your extended infrastructure, architecture, and/or implementation.
- Analyses and discuss the capabilities of the project.
- Present evidence in support of your analysis.
- (PG) Describe the methodology of your experimental evaluation, and present the result of your evaluation. This presents a critical quantitative or qualitative evaluation of the strengths and limitations of your work.
- Includes a full list of references to existing software and relevant literature.

You should structure this report similar to a research paper, to contain sections such as:

- Introduction
- Related Work (Existing Methods)
- Methodology
- Results
- Analysis & Evaluation
- Conclusion

You should including data tables and figures that support the written text. Figures and tables should be formatted to be fully legible at default 100% PDF zoom level, and legible when printed in black-and-white.

The format of your report should be:

- No more than 10 pages, including figures, tables and references.
- Single Column
- No less than 1.5cm margins
- No less than 11pt font.

7 Submission

Follow the instructions on **Canvas** to complete your submission for the Assignment.

Assessment declaration: When you submit work electronically, you agree to the assessment declaration.

7.1 Help during Weeks 13-15

After Week 12, there are no formally scheduled classes. Please note that limited help will be available during this time frame, and any in-person help will need to be conducted by a pre-arranged appointment. Therefore, you should ensure that you discuss all necessary and important issues/question during your progress update.

7.2 Late Submissions & Extensions

A penalty of 10% per day is applied to late submissions up to 5 business days, after which you will lose ALL the assignment marks. Extensions will be given only in exceptional cases; refer to Special Consideration process.

8 Marking guidelines

The marks are divided as follows:

- Progress Update
 - Demonstration: 10/50
- Final Demonstration
 - Demonstration of Implementation: 10/50
 - Demonstration of Evaluation and Results: 5/50
 - Individual Contribution: 5/50
- Report (40%)
 - Description of Methodology: 5/50
 - Analysis and Evaluation: 10/50
 - Writing and Referencing style: 5/50

The detailed breakdown of this marking guidelines is provided on the rubric linked on Canvas.

9 Academic integrity and plagiarism (standard warning)

Academic integrity is about the honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge and ideas. You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e. directly copied), summarised, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods
- Provided a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites. If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct. Plagiarism covers a variety of inappropriate behaviours, including:

- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students

For further information on our policies and procedures, please refer to the RMIT Academic Integrity Website.

The penalty for plagiarised assignments include zero marks for that assignment, or failure for this course. Please keep in mind that RMIT University uses plagiarism detection software.